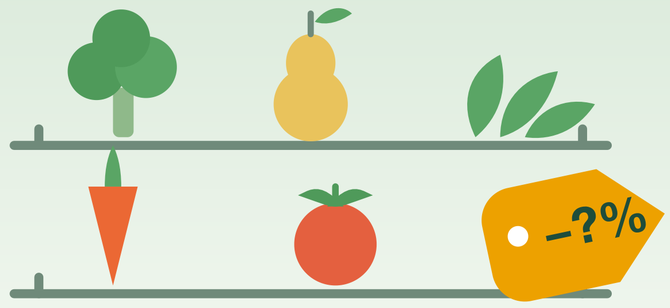
HKU MAIB 7002 · GROUP Q · MACHINE LEARNING FOR PRICING

Cut the price, or trust the shelf?

We taught a model to predict what a perishable will sell tomorrow at any discount. Then we discovered it would discount everything, and had to teach it to read the shelf first.

Aashish Omprakash Pareek · Yuanyuan Yin · Lingyu Chen · Dingdong FreshRetailNet-50K, a public dataset of 50,000 store-product histories



6 pm · the shelf is still half full · what is tomorrow's price?

01 A decision every evening, for every product

6 pm, the shelf is half full

Tomorrow's discount must be set tonight, before anyone knows how much will sell.

Two ways to lose

Hold the price and stock may be thrown away. Cut it and margin goes on sales that would have happened anyway.

Our question

For one product, in one store, on one day: how much sells tomorrow at each discount, and should we discount at all?

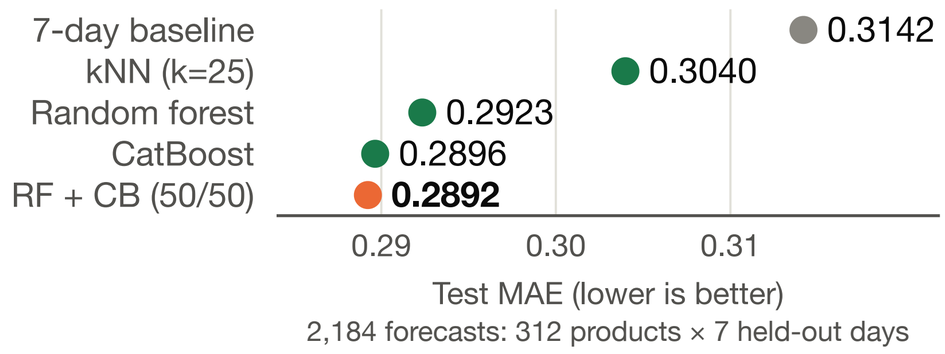
02 The raw material

312
product histories in five stores, picked because their discounts vary

28,080
store-product-days; one unseen week of 2,184 kept for the test

45%
of days have an empty-shelf hour, a clue we later put to work

03 Forecasting results



2,184 forecasts: 312 products × 7 held-out days

7.9% lower MAE than the 7-day average with the 50/50 random forest + CatBoost blend; squared error 22% lower

What we learned: every model beat the no-ML average, the three tree models tie within noise, and choosing the inputs mattered more than any tuning dial. Dropping the weather was the single biggest gain.

04 Model choices

kNN · 25 neighbours

Averages the 25 most similar past days; dropping weather cut its error from 0.327 to 0.312.

CatBoost · 500 trees

Depth 5, learning rate 0.03: small and slow on purpose, because boosting over-fits.

Random forest · 300 trees

Depth 10, 3 rows per leaf; all 16 settings beat the baseline every week.

RF + CatBoost · 50/50 — FINAL MODEL

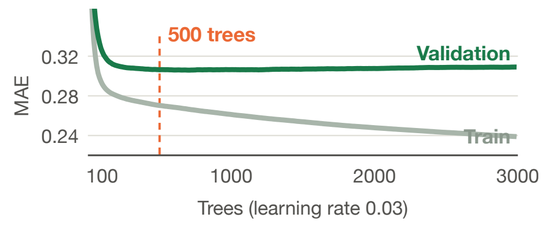
Averages both forecasts; errors correlate at 0.97, so the gain is small but consistent.

Random forest: validation MAE across 16 settings

	3	5	10	20
10	0.3024	0.3027	0.3035	0.3052
15	0.3030	0.3031	0.3032	0.3049
20	0.3043	0.3036	0.3034	0.3048
None	0.3044	0.3034	0.3031	0.3051

rows: max depth · columns: min rows per leaf · best outlined

CatBoost: error as trees are added



05 The twist: a good forecaster is a reckless pricer

Ask the model "what if I discount?" and it answers: deeper is always better. At a 0.6 rate it predicts 2.3× the sales.



It learned that from clearance days, which are fewer than 1% of the data. Left alone it would discount on 98% of days.

So we put the shelf before the model. Three stock rules decide first; the model only chooses among discounts the product has actually run.

Sold out today?

Then no discount tomorrow: the shelf is clear.

A week with no sell-out?

Then a discount is required; the model picks how deep.

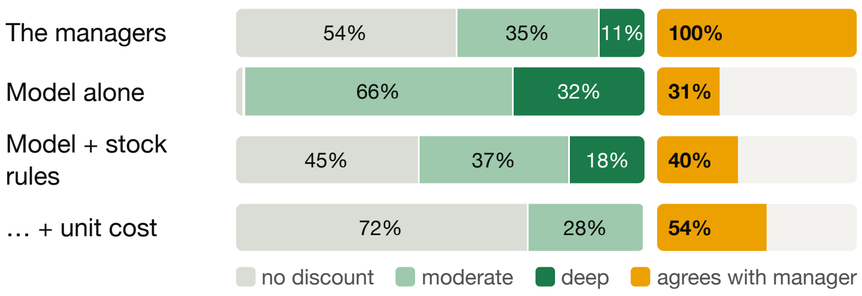
Otherwise ask the model

Only about discounts this product has actually run.

Choose

Highest predicted value, net of unit cost, if the gain is worth it.

What each policy recommends on the test week, and how often it lands in the same band as the real managers.



Stock rules decide 53% of days before the model is asked. Agreement rises from 31% to 40%, and to 54% once a unit cost makes deep discounts cost something.

06 The challenges we found most interesting

The shelf lies

On 45% of days the shelf was empty for part of the day, so recorded sales are lower than real demand. The model learns from censored numbers.

Sales without units or prices

The data is normalised: 1.2 is not 1.2 kilograms or dollars. "Value" is sales × discount rate, a proxy, so profit cannot be computed.

Managers are not random

Discounts were chosen on days managers expected demand. Part of the lift the model sees is their judgement, not the price cut. Correlation, not cause.

No counterfactual

We only ever see what sold at the price the manager chose. Whether our policy sells more or wastes less needs a real experiment.

Weather made things worse

0.327 → 0.312

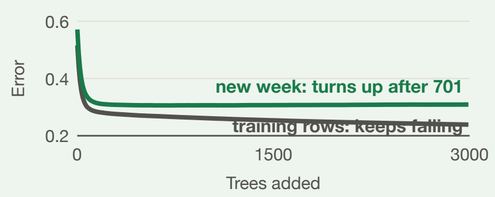
Removing rain, temperature and humidity cut the error more than any amount of tuning did.

Deep discounts are the hard days



Error by tomorrow's discount: every model under-predicts the spike on exactly the days a pricing tool cares about.

Boosting over-fits in plain sight



Training error keeps falling; error on a new week turns up after 701 trees.

One city, one week, one guess at shelf life

Five stores in one city and a single test week that has been scored for four models. The "no sell-out for a week means old stock" rule assumes a shelf life the data never states.

Data: Dingdong-Inc/FreshRetailNet-50K on Hugging Face, CC BY 4.0 · scikit-learn and CatBoost · Errors are mean absolute error in normalised sales units

Full notebook, code and Docker image: github.com/Lisayinyy/MAIB_7002, branch pricing-ml-ap